26900397 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCPES300 - PHYSIOLOGY OF EXERCISE

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. What major physiological discovery was made by William Harvey in 1628? [K] / [CO1]

   a). The structure of DNA | b). The continuous circulation of blood | c). The concept of homeostasis | d). The role of oxygen in respiration

2. Because the body stores very little raw ATP, what must cells constantly do? [U] / [CO2]

   a). Rely entirely on fat stores | b). Stop moving to conserve energy | c). Recycle millions of ADP molecules back into ATP | d). Produce energy without ATP

3. Which of the following is NOT an adaptation of skeletal muscle to regular exercise? [E] / [CO3]

   a). Muscle hypertrophy | b). Increased mitochondria | c). Decreased capillary density | d). Increased muscle endurance

4. Why are there numerous mitochondria in the presynaptic terminal? [An] / [CO3]

a). To store calcium | b). To provide the energy needed for neurotransmitter release | c). To bind acetylcholine | d). To break down enzymes

5. Skinfold calipers measure: [K] / [CO1, CO6]

a).Bone density | b).Subcutaneous fat thickness | c).Visceral fat | d).Muscle mass

6. Body fat percentage >25% in men is classified as: [U] / [CO1, CO6]

a).Underweight | b).Normal | c).Overfat | d).Obese

7. How does trained respiratory efficiency affect breathing at rest? [U] / [CO3]

a). Breathing is rapid and shallow | b). Breathing rate is reduced because each breath is deeper and more efficient | c). Breathing stops | d). It has no effect

8. What causes the initial feeling of fatigue before a 'second wind' occurs? [An] / [CO3]

a). Excess oxygen in the blood | b). An initial oxygen deficit as the cardiovascular system takes time to adjust | c). Low heart rate | d). High blood sugar

9. Which exercise approach is best for long-term BP control? [E] / [CO1, CO2, CO6]

a). Occasional maximal exercise | b). Complete inactivity | c). Regular moderate exercise | d). Sprint training only

10. Which metabolic condition is commonly associated with PCOD? [K] / [CO1, CO2, CO6]

a). Hypotension | b). Insulin resistance | c). Anaemia | d). Asthma

[1x10 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

11. Describe the molecular structure of ATP. [K] / [CO2]

12. What happens to the pyruvic acid produced during glycolysis if no oxygen is present? [U] / [CO2]

13. Give an example of heat loss via radiation. [A] / [CO4]

14. Name two hormones that significantly influence baseline metabolism. [K] / [CO2]

15. Define fat mass and explain its role in the body. [K] / [CO1, CO6]

16. Explain why exercise prescription should be individualized for people with chronic diseases. [U] / [CO1, CO2, CO6]

17. Justify the need to consider individual variation when studying sex differences. [E] / [CO7]

[2x5 = 10]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

18. Outline five current sub-disciplines of modern physiology and explain their primary focus. [U] / [CO1]

19. Summarize the holistic importance of metabolism in the human body, detailing how it supports growth, tissue repair, body temperature regulation, and overall health. [U] / [CO1]

20. Describe the structural components of the neuromuscular junction (NMJ) and explain how the destruction of acetylcholine receptors on the motor end plate would affect muscle contraction. [An] / [CO3]

21. Compare lean body mass adaptations to resistance training versus endurance training. [An] / [CO1, CO6, CO7]

22. Explain how the principles of progression and overload can be applied to develop strength and muscular fitness in male and female students. [A] / [CO7]

23. Explain aerobic exercise prescription for inactive older adults. [A] / [CO1, CO2, CO6]

[5x4 = 20]

Part D

Essay Questions

Answer any **2** questions

Each question carries **15** marks

24. Catabolism vs. Anabolism: Define human metabolism and clearly distinguish between its two primary phases: Catabolism and Anabolism, providing examples for each. [U] / [CO2]

25. Acute Cardiovascular Responses: Analyze the short-term (acute) cardiovascular responses to a single bout of exercise, specifically explaining changes in heart rate, stroke volume, and cardiac output. [An] / [CO3]

26. Evaluate the relationship between BMI and morbidity and mortality risk. Discuss the evidence for BMI as a predictor of health outcomes. [E] / [CO1, CO6]

27. Critically assess the influence of food habits and physical inactivity on chronic bronchitis. [E] / [CO1, CO2, CO6]

[15x2 = 30]